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A simplified model of soakaway infiltration interaction with a shallow groundwater table

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SUMMARY

This paper presents a new and simplified modeling concept for soakaway infiltration in the presence of a shallow groundwater table, including representation of the local groundwater mound and its effects on the infiltration rate. The soil moisture retention curve is used to represent the influence of the mound on infiltration rates. The model is intended to be used in situations when distributed urban drainage models with soakaways or similar infiltration devices are coupled to distributed groundwater flow models. With this new modeling concept, the local mounding from small-scale infiltration systems, and its effects on the infiltration rate, can be represented even if the spatial resolution of the groundwater flow model is coarser than the extent of the mound. The new model has been run for a number of scenarios and soil parameters, and the results compared to the output from a two-dimensional unsaturated/saturated flow model based on Richard's equation. The comparison shows that soakaway emptying times calculated by the new model are on average 13% higher than the emptying times of the two-dimensional model. The deviation is smaller for scenarios including a shallow groundwater table, only around 5% on average. Groundwater mounding depths are underestimated close to the soakaway and overestimated far from the soakaway. At 5 m distance from the soakaway center, the average deviation is 18% for the shallow groundwater scenarios. Mass balance errors were checked and found to be below 1% for all scenarios at all times during the simulation period. The extra uncertainty introduced by this new model is compensated for by the reduction in runtime; it is on average 600 times faster than the two-dimensional model. Furthermore, the new model is based on the same input parameters as the two-dimensional model and thus requires no extra calibration. The new modeling concept is therefore a useful tool for simulating small-scale stormwater infiltration in the presence of a shallow groundwater table with distributed models on larger scales.

1. Introduction

Soakaways, also known as infiltration trenches, are commonly used to infiltrate stormwater runoff from e.g. roofs, parking areas and roads (Revitt et al., 2003; CIRIA, 2007). If properly designed, they can lead to substantial decrease of stormwater load on the sewer network. However, infiltration of stormwater can be challenging. Polluted stormwater is a potential hazard to groundwater resources, land use may limit the space available for infiltration devices, and geological and hydrogeological restrictions such as low permeability soils and/or a shallow groundwater table may decrease the efficiency of these devices (Roldin et al., 2012b; Bergman et al., 2011). If frequently used in an area, they may also induce rising groundwater tables, groundwater mounds and surface flooding that can create a potential flood risk for houses and basements (Antia, 2008; Endreny and Collins, 2009). It is therefore important to be able to accurately predict the effects of soakaways and similar infiltration structures on both sewer flows and groundwater flows, in order to promote a more extensive use of such structures (Dietz, 2007). This includes coping with the bi-directional interaction between infiltration and groundwater that will occur unless the groundwater level is very deep (Bouwer, 2002).

The effect of soakaway infiltration on local groundwater levels, and the formation of mounds for various conditions, has been extensively described and modeled in literature (e.g. Guo, 1998; Bouwer, 2002; Endreny and Collins, 2009; Thompson et al., 2010; Carleton, 2010; Machusick et al., 2011; Maimone et al., 2011). Increased infiltration and groundwater recharge through soakaways and similar devices has also been shown to affect regional groundwater levels (Ku et al., 1992; Göbel et al., 2004; Markussen et al., 2004; Jeppesen, 2010; Maimone et al., 2011). The general

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conclusion is that infiltration-based stormwater systems can indeed lead to a substantial increase in the long term average water table level, as well as significant groundwater mounds, and that it is important to consider groundwater levels when planning for stormwater infiltration. Furthermore, regional groundwater levels and local groundwater mounds are known to affect the infiltration rate from soakaways (Duchene et al., 1994; Bouwer, 2002), because the infiltration rate decreases as the water table approaches the soakaway. The soakaway-groundwater system can thus be described as a fully coupled system where the infiltration rate and the groundwater level influence each other. This system is modeled in various ways in existing studies.

On a small scale, where only one or a few soakaways are modeled, a common approach is using a numerical unsaturated/saturated two-dimensional or 3D flow model based on Richards' equation (Richards, 1931), which is able to account for this two-way interaction (e.g. Duchene et al., 1994; Maimone et al., 2011). Moreover, several studies have examined the interaction between neighboring soakaways where the mounds are interconnected (Endreny and Collins, 2009; Maimone et al., 2011).

On a regional scale, the coupling between soakaway infiltration and groundwater has been modeled by e.g. Göbel et al. (2004), Markussen et al. (2004), Jeppesen (2010) and Maimone et al. (2011). A frequently used procedure is to first evaluate the infiltration rates on a local scale with a finely discretized two-dimensional or 3D model where the effects of local mounding can be accounted for, and then upscale this result to a regional scale by using the resulting infiltration time series as input to a regional groundwater model (e.g. Göbel et al., 2004; Maimone et al., 2011). Others used effective infiltration rates in coarse grid groundwater models to represent the infiltration from soakaways so that only regional groundwater changes (and not local mounds) affect the infiltration rate (e.g. Jeppesen, 2010), or simply that the infiltration rate is not affected by groundwater levels (e.g. Ku et al., 1992).

Soakaways also interact with other urban water systems, in particular the sewer network. In existing studies of this integrated system, sequential modeling has generally been employed. First the infiltration has been quantified with an urban drainage model capable of simulating stormwater infiltration systems, and subsequently this infiltration hydrograph has been used as an input to a groundwater model to assess the effects on the groundwater levels (Markussen et al., 2004; Endreny and Collins, 2009). An alternative method has been to first estimate the maximum allowed increase in groundwater recharge based on a hydrological model, then calculate infiltration based on this knowledge and finally evaluate the effects on the sewer flows (Roldin et al., 2012b). However, the use of sequential modeling will not be able to account for the full coupling between infiltration systems and groundwater (Schmitt and Huber, 2006). This may be accurate enough if the initial groundwater level is deep, but if the groundwater level and mounding starts to approach the infiltration system, the infiltration rate will inevitably be affected (Bouwer, 2002). Furthermore an integrated modeling approach of sewer flows, stormwater infiltration and groundwater flows will require consideration of the variety of temporal and spatial scales involved. In the studies where groundwater mounding and stormwater infiltration has been investigated with a 2- or 3-dimensional flow model, the grid cells have been sized to be able to capture the local mounding effect, with an element size usually in the range of one or a few metres (Göbel et al., 2004; Markussen et al., 2004; Endreny and Collins, 2009; Thompson et al., 2010; Maimone et al., 2011). However, if the effects of stormwater infiltration are to be assessed on a larger scale, this fine grid will quickly lead to unacceptable computational time and effort.

A fully integrated approach for soakaways would involve simultaneous modeling of urban drainage models and groundwater models, where local effects from soakaways can be represented and accounted for. In this paper we present a new concept for this, where the coupling between soakaways and local groundwater mounds are represented by simplified analytical expressions in an additional soakaway model component that can be coupled to urban drainage and groundwater models. With our method, effects of small-scale infiltration structures like soakaways can be captured and represented even if the urban drainage and groundwater models are run on a much coarser spatial scale compared to the size of a standard soakaway or groundwater mound.

The aim of this paper is to develop a new infiltration model for soakways and similar structures that can be used to estimate the effects of a shallow groundwater table on the infiltration rate, as well as the local groundwater mounding effects. The new model is intended to be used when integrating physically distributed urban drainage models with grid-based groundwater flow models. The new approach represents the interaction between the groundwater table and the soakaway by simplified analytical equations in the soakaway component, based on the soil moisture retention curve and the Hantush equation for groundwater mound (Hantush, 1967). With our method, the resolution of the groundwater model can be much coarser than the size of the local mound extent, but the effect of the local mound on the infiltration is still accounted for.

2. Methodology

This section describes the newly developed model, as well as a two-dimensional unsaturated/saturated flow model that was used to validate the results from the new model. We will also evaluate the suitability of the two models for use in an integrated modeling environment with respect to computational time and data requirements.

2.1. Model overview and common features

Fig. 1 shows a schematic overview of the two models that are used in this study. Both models assume that the soakaway is "infinitely long" with symmetrical conditions around the center of the soakaway, and calculations are therefore done for half the width of the soakaway, with volumes given as "m^3/m soakaway length". Parameters that have been used in one or both models are listed and explained in Table 1.

The mass balance of the soakaway is identical in both models and is given by:

$$B \cdot \varphi \cdot \frac{dSH}{dt} - Q_{in} + Q_{out} = 0 \tag{1}$$

where B is the half width of the soakaway [L], φ is the porosity of soakaway filling material [–], SH is the soakaway water level [L], Q_{in} and Q_{out} are the inflow and outflow per unit length of soakaway [L^2T^{-1}] respectively and t is time [T].

The Van Genuchten (1980) equation describes the soil moisture content (θ) as a function of the head (h):

$$\theta = \theta_r + \frac{\theta_s - \theta_r}{\left(1 + (\alpha|h|)^n\right)^m} \quad \alpha > 0; \quad n > 1 \tag{2}$$

where θ_r is the residual moisture content [–], θ_s is the saturated moisture content [–], h the pressure head [L], α [L^{-1}], n [–] and m [–] are the soil specific parameters.

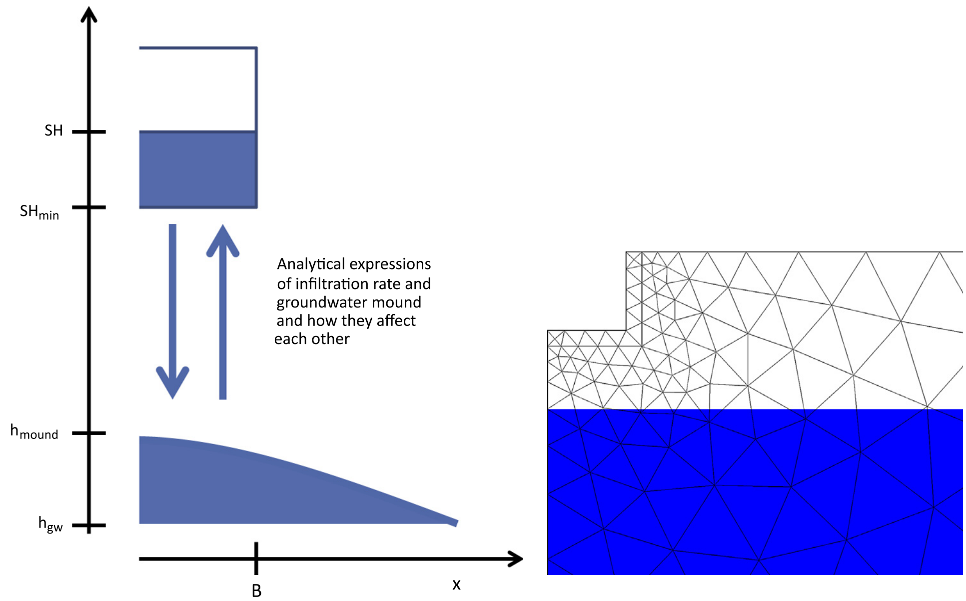

Fig. 1. Left – new modeling concept based on mass balance, approximation of the infiltration rate with the water retention curve and solving the Hantush equation. Right – two-dimensional unsaturated/saturated flow model based Richards equation (only part of the modeling domain is shown); the blue area indicates the saturated zone. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Model parameters.

Symbol	Description	Value	Unit
α	Van Genuchten parameter	See Table 3	m^{-1}
B	Half width of the soakaway	1	m
f	Infiltration rate per m^2 wetted soakaway area		m/s
h	Pressure head	–	m
h_{gw}	Initial groundwater level above datum	See Table 3	m
h_{mound}	Groundwater mound level, above datum	–	m
K_s	Saturated hydraulic conductivity	See Table 3	m/s
L	Extent of soil domain in x-direction	50	m
n	Van Genuchten parameter	See Table 3	–
m	Van Genuchten parameter	$1 - 1/n$	–
φ	Porosity of soakaway filling material	≈ 1	–
$q_{\max}$	Maximum infiltration rate per area unit	–	m/s
Q_{2grw}	Inflow to groundwater from soil		m^2/s
$Q_{DrainMax}$	Drainage rate from center of groundwater mound	–	m^2/s
Q_{in}	Inflow to soakaway	Eq. (18)	m^2/s
Q_{out}	Infiltration from soakaway to soil		m^2/s
SH	Soakaway water level above datum	–	m
SH_{max}	Soakaway top level above datum	20	m
SH_{min}	Soakaway bottom level above datum	19	m
S_s	Specific storage	10^{-4}	m^{-1}
S_w	Relative saturation	–	–
t	Time	–	s
θ_s	Saturated moisture content of soil	See Table 3	–
θ_r	Residual moisture content of soil	See Table 3	–
θ_i	Initial moisture content of soil at soakaway bottom	–	–

2.2. A coupled infiltration and groundwater mounding model

2.2.1. Infiltration rate model

The infiltration rate from a soakaway or similar structure will be affected by the presence of a shallow groundwater table (Bouwer, 2002). One way to account for this is to use the soil moisture retention curve (Eq. (2)) to adjust the infiltration rate. A similar idea was employed by Crosbie et al. (2005) to adjust water level changes as a result of infiltration.

$$f = \frac{\theta_s - \theta_{(\mathrm{h}=\Delta h)}}{\theta_s - \theta_i} \cdot q_{\max} \tag{3}$$

where θ_s is the saturated moisture content [–], θ_t is the moisture content at the bottom of the soakaway [–], $\theta_{(h=\Delta h)}$ is the moisture content [–] calculated from Eq. (2) at the point $h = \Delta h$, Δh [L] is the distance between the soakaway bottom and the groundwater level below the soakaway, here defined as the top of the groundwater mound, i.e. $\Delta h = SH_{\min}–h_{mound}(x = 0)$ and $q_{\max}$ [LT^{-1}] is the maximum infiltration rate per area unit which is assumed to be equal to the saturated hydraulic conductivity (i.e. unit-gradient Darcy flow). This equation assumes hydrostatic pressure distribution in the groundwater mound with the moisture content in between the bottom of the soakaway and the top of the mound calculated using Eq. (2). Setting $q_{\max}$ to equal the saturated hydraulic conductivity

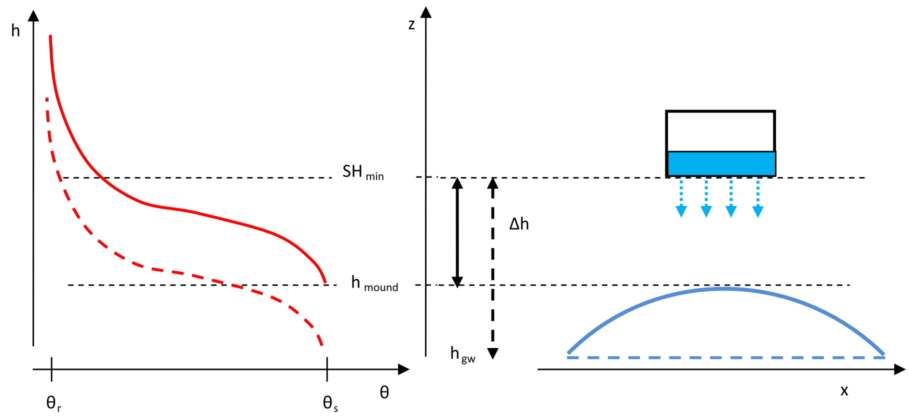

Fig. 2. Illustration of how mounding in combination with an estimated soil water retention curve is assumed to affect the moisture content under the soakaway (adopted from Crosbie et al., 2005). Blue lines are groundwater levels and red curves soil water retention curves. Dashed line marks initial conditions (no mounding) and solid line marks conditions with mounding effect. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

implies that suction effects as well as effects of decreased hydraulic conductivity in unsaturated soils are ignored. The assumption that these effects are negligible can be reasonable in situations where the soil is frequently wetted and seldom very dry, which is often the case around a soakaway, in particular if the groundwater table is near. This unit-gradient approach has been found to work well in previous soakaway studies (e.g. Warnaars et al., 1999).

Eq. (3) shows that when the mounding height is low (mound height close to initial groundwater table), the infiltration rate is close to its maximum $q_{\max}$. As the top of the mound approaches the soakaway, the infiltration rate will decrease at a rate depending on the soil moisture retention curve, to finally become 0 when the groundwater level reaches the soakaway water level. See Fig. 2 for details.

Infiltration is assumed to occur through the entire wetted area of the soakaway, and the infiltration rate per unit area (f) is therefore multiplied by the wetted area of the soakaway (or, in this case, the wetted perimeter since the model is only 2-dimensional) to obtain the total infiltration discharge from the soakaway (Eq. (4)). The infiltration equation (Eq. (3)) tends to 0 as the groundwater mound approaches the soakaway bottom. In reality, if the groundwater mound reaches the soakaway bottom or above, the infiltration rate will equal the mound drainage rate. Thus, an extra condition is applied to Q_{out} to set it to be the groundwater mound drainage rate if the top of the mound has reached the soakaway bottom.

$$Q_{out} = \begin{cases} 0 & SH \le SH_{\min} \\ f(B + SH - SH_{\min}) & SH > h_{mound}; SH > SH_{\min} \\ \max(f(B + SH - SH_{\min}), Q_{DrainMax}) & SH \leqslant h_{mound}; SH > SH_{\min} \end{cases} \tag{4}$$

where B is the half width of the soakaway [L], SH is the soakaway water level above datum [L], $SH_{\min}$ is the bottom level of the soakaway above datum [L], f is the infiltration rate per unit area [LT^{-1}] and $Q_{DrainMax}$ [L^2T^{-1}] is the drainage rate from the center of the groundwater mound calculated from the groundwater mounding function (Eq. (8)) and explained in Section 2.2.2., Eq. (11).

Eqs. (2)–(4) are thus the description of how the groundwater level affects the infiltration from the soakaway. The interaction is fully coupled (bidirectional) with the local groundwater level (h_{mound}) depending on the infiltration rate from the soakaway and vice versa.

2.2.2. Groundwater mounding model

The local groundwater mounding beneath the soakaway is described by the well known solution of Hantush (1967) for a rectangular recharge area in the x–y plane (dimension $A{\cdot}B$) with constant infiltration rate w:

$$h_{mound}^2 - h_{gw}^2 = \left(\frac{w}{2K_s}\right)(vt)\left\{S\left(\frac{B+x}{\sqrt{4vt}}, \frac{A+y}{\sqrt{4vt}}\right) + S\left(\frac{B+x}{\sqrt{4vt}}, \frac{A-y}{\sqrt{4vt}}\right) + S\left(\frac{B-x}{\sqrt{4vt}}, \frac{A+y}{\sqrt{4vt}}\right) + S\left(\frac{B-x}{\sqrt{4vt}}, \frac{A-y}{\sqrt{4vt}}\right)\right\} \tag{5}$$

where h_{mound} and h_{gw} are the height of the mounding and the groundwater table above datum [L] respectively. K_s is the saturated hydraulic conductivity [LT^{-1}]. The function S is defined by:

$$S(a,b) = \int_0^1 erf\left(\frac{a}{\sqrt{\tau}}\right) erf\left(\frac{b}{\sqrt{\tau}}\right) d\tau \tag{6}$$

and

$$v = \frac{K_s}{S_y} \frac{h_{gw} + h_{mound}}{2} \tag{7}$$

The specific yield S_y (the drainable porosity, as defined by Hilberts et al., 2005) is estimated to equal the average soil moisture deficit ($\Delta\theta = \theta_s - \theta$) in the unsaturated zone between the soakaway bottom and the groundwater table. The value of θ is evaluated at three points – at the soakaway bottom, halfway between the soakaway bottom and the groundwater table, and at the groundwater table – and the average soil moisture deficit is calculated from these three values.

A is half the length of the soakaway, which is assumed to be infinite, and since $erf(\infty) = 1$, Eq. (5) reduces to:

$$h_{mound}^2 - h_{gw}^2 = Z(w,t) = \left(\frac{w}{K_s}\right)(vt)\left\{S\left(\frac{B+x}{\sqrt{4vt}}, \infty\right) + S\left(\frac{B-x}{\sqrt{4vt}}, \infty\right)\right\} \tag{8}$$

The original Hantush equation was developed for constant recharge rate, but since the equation is linear in h_{mound}^2, additional terms can be added to the right hand side of Eq. (8) to account for the effect of a variable recharge rate, assuming that the recharge rate can be described as a series of piecewise constant recharge rates (Hantush, 1967). If for instance the recharge rate is as shown in the following equation:

$$w = \begin{cases} w_1 & 0 \le t \le t_1 \\ w_2 & t_1 \le t < t_2 \\ w_3 & t_2 \le t < t_{end} \end{cases} \tag{9}$$

where w_1, w_2 and w_3 are different constants, then the resulting mounding equation will be (Eq. (10)):

$$h_{mound}^2 - h_{gw}^2 = Z(w_1, t_{end}) + Z(w_2 - w_1, t_{end} - t_1) + Z(w_3 - w_2, t_{end} - t_2) \quad (10)$$

where $Z(w,t)$ is the function on the right hand side of Eq. (8).

$Q_{DrainMax}$ (which is used in the infiltration equation, Eq. (4)) is calculated by inverting Eq. (8), setting $x = 0$ and t = current simulation time, and relating $Q_{DrainMax}$ to the recharge rate w (which also equals the drainage rate from the center of the mound):

$$Q_{DrainMax} = Bw = BK_s \frac{h_{mound}^2 - h_{gw}^2}{2(vt)\left\{1 - i^2 \cdot S\left(\frac{B}{\sqrt{4vt}}\right)\right\}} \quad (11)$$

where B is the half width of the soakaway [L] and term $i^2{\cdot}S$ is the second repeated integral of the error function as discussed in Carslaw and Jaeger (1959) and used in Hantush (1967). This term can also be found in any versions of the Handbook of Mathematical Functions (Abramowitz and Stegun, 1964; Olver et al., 2010).

2.2.3. Lag time

For situations with low permeability soils and/or a distant groundwater table, the infiltrated water will not immediately recharge the groundwater and contribute to a local mounding effect. To account for this effect, a lag time is introduced as shown in the following equation:

$$\begin{aligned} Q_{2grw}(t) &= 0 \qquad & t < t_{lag} \\ Q_{2grw}(t) &= Q_{out}(t - t_{lag}) & t \geqslant t_{lag} \end{aligned} \quad (12)$$

The lag time is estimated by simply dividing the distance between the soakaway bottom and the initial groundwater table by the pore velocity of the water. The pore velocity is estimated to be the saturated hydraulic conductivity divided by the specific yield:

$$t_{lag} = \frac{SH_{\min} - h_{gw}}{K_s/S_y} \quad (13)$$

2.2.4. Numerical implementation

The soakaway water balance is solved using a forward Euler timestepping method with Δt = 600 s. To simplify and speed up the calculations, the infiltration model and groundwater model are solved sequentially in each time step; first the infiltration model is solved with groundwater information from the previous time step, and then the groundwater model is solved with infiltration information from the current time step. The mass balance was tracked throughout the entire simulation, and additional runs were made with Δt = 60 s and Δt = 3600 s to ensure that numerical errors with the selected time step length are small.

An analytical solution to the Hantush equation for a constant recharge rate is employed, based on a finite Fourier sine transform series (Rao and Sarma, 1983). Other approaches could include those of Carleton (2010) who proposed a method based on numerical integration of the Hantush equation, and Trefry (1998) and Mamedov and Ekenoglu (2006) who proposed a method to find the value of Hantush S function by using the related Hantush M function. Since the recharge rate is not constant throughout the simulation period, it has been averaged to maximum five constant terms according to Eq. (14) (t^* is the current time in hours in the simulation):

$$Q_{2grw} = \begin{cases} Q_{2grw,1} & t^* - 24 < t \leq t^* \\ Q_{2grw,2} & t^* - 72 < t \leq t^* - 24 \\ Q_{2grw,3} & t^* - 144 < t \leqslant t^* - 72 \\ Q_{2grw,4} & t^* - 240 < t \leq t^* - 144 \\ Q_{2grw,5} & t \leq t^* - 240 \end{cases} \quad (14)$$

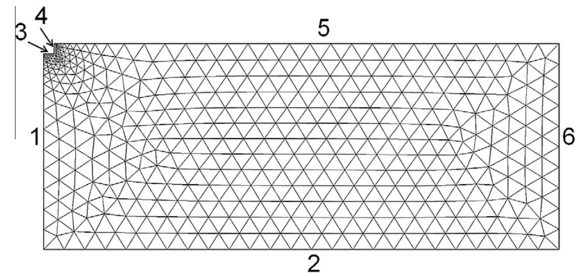

Fig. 3. Geometry of the two-dimensional model with boundary numbers and finite element mesh.

$Q_{2grw,1}$ is the average recharge rate for the 24 h before the current simulation time, $Q_{2grw,2}$ the average recharge rate for the 48 h before that, etc. This means that the Hantush equation is expanded to a maximum of five terms on the right hand side of Eq. (8) as explained in Section 2.2.2.

The function v (Eq. (7)) is calculated using the h_{mound} value from the previous timestep, to avoid excessive use of implicit equation solutions.

The model is set up in Matlab, but to facilitate coupling with various hydrological models, the soakaway model can also be set up as an OpenMI linkable component (Gregersen et al., 2007) which would greatly facilitate simultaneous modeling with OpenMI compliant urban drainage and groundwater modeling software.

2.3. Two-dimensional unsaturated/saturated flow model

The two-dimensional soil and groundwater flow model solves Richards' equation (Richards, 1931) which written in a form that can be used for both saturated and unsaturated conditions. The model geometry is shown in Fig. 3 and the flow equation in the following equation:

$$(C + S_s \cdot S_w)\frac{\partial h}{\partial t} - \left(\nabla \cdot (K \cdot \nabla h) + \frac{\partial K}{\partial z}\right) = 0 \quad (15)$$

C is the specific moisture capacity ($\frac{\partial \theta}{\partial h}$) and S_w is the relative saturation (Eq. (16)):

$$S_w = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad (16)$$

and K is the hydraulic conductivity as a function of pressure head (Van Genuchten, 1980):

$$K = K_s\left(\frac{\left[1 - (\alpha|h|)^{n-1}(1 + (\alpha|h|)^n)^{-m}\right]^2}{\left[1 + (\alpha|h|)^n\right]^{m/2}}\right) \quad (17)$$

The two-dimensional flow model is coupled to the soakaway mass balance model (Eq. (1)) which provides the boundary conditions for boundaries 3 and 4 (see Fig. 3 for boundary location).

The model is solved using COMSOL Multiphysics which employs a finite element solver. Boundary conditions are listed in Table 2. To increase numerical stability, a 0.2 m thick zone was added around the soakaway where the initial head was set to −0.5 m, i.e. close to saturation. This corresponds to a situation where the soil closest to the soakaway is initially wet – which is considered to be a realistic assumption.

2.4. Scenarios and input parameter values

2.4.1. Base scenarios

Six base scenarios were initially defined – three different soil types (sandy clay loam silt loam and clay loam), all with two different initial groundwater levels (1 m and 5 m below the soakaway).

Table 2
Boundary conditions and initial condition for the two-dimensional unsaturated/saturated flow model.

Boundary condition (BC) or initial condition (IC)	Description
BC 1, 2 and 5	**Zero flux**. No flow through upper boundary (ground), lower boundary (assumed impervious layer) or the left boundary (for symmetry reasons)
BC 3	***h* = *SH***. The pressure head is equal to the soakaway water depth
BC 4	***h* = *SH* + *SH*$_{\min}$ − *z***. The pressure head is hydrostatic relative to the value given on boundary 3
BC 6	$h = \begin{cases} h_{gw} - z \text{for } z < (h_{gw} - h_{\min}) \\ h_{\min} \text{for } z \geqslant (h_{gw} - h_{\min}) \end{cases}$. A hydrostatic condition but with a minimum pressure head of $h_{\min}$, which is the pressure head at a water content of 0.2. See Table 3 for values of $h_{\min}$
IC	$h(t = 0) = h$ (BC 6). Hydrostatic conditions as described for boundary 6

Table 3
Base scenario parameters. Saturated hydraulic conductivity and soil moisture retention parameters for the three soil types used in the study (from Carsel and Parrish, 1988). The $h_{\min}$ parameter refers to the suction head corresponding to $\theta = 0.2$. The two initial groundwater levels used in the base scenarios are also listed.

Parameter	Sandy clay loam	Silt loam	Clay loam	Unit
K_s	3.64×10^{-6}	1.25×10^{-6}	7.22×10^{-7}	m/s
θ_s	0.39	0.45	0.41	–
θ_r	0.1	0.067	0.095	–
α	5.9	2.0	1.9	m^{-1}
n	1.48	1.41	1.31	–
$h_{\min}$	−1.52	−6.47	−18.05	m
h_{gw}	14 (“deep”)	14 (“deep”)	14 (“deep”)	m
or	18 (“shallow”)	18 (“shallow”)	18 (“shallow”)	m

Table 4
Alternative scenarios and parameter variation.

Scenario	h_{gw} (m)	K_s (m/s)
Base	18	1.25×10^{-6}
Deep groundwater, GWD	14	1.25×10^{-6}
Intermediate groundwater, GWI	16	1.25×10^{-6}
Low conductivity, Klow	18	1.25×10^{-7}
High conductivity, Khigh	18	1.25×10^{-5}

The two groundwater scenarios are named “shallow” (1 m below) and “deep” (5 m below). The distance from the soakaway down to the bottom of the confining layer is 19 m and the initial unconfined aquifer thickness is either 18 m (“shallow”) or 14 m (“deep”).

Table 3 lists the saturated hydraulic conductivity and soil moisture retention parameters for the three soil types that have been used in the study – sandy clay loam, silt loam and clay loam (from Carsel and Parrish, 1988).

The inflow hydrograph Q_{in} is the same for all base scenarios and is given by Eq. (18). It is a simple inflow hydrograph describing a constant inflow to the soakaway over a 2-h period:

$$Q_{in}\left(\frac{m^2}{s}\right) = \begin{cases} \frac{1}{7200} & t < 7200s \\ 0 & t \geq 7200s \end{cases} \tag{18}$$

This corresponds to a 2 h rainfall event of 20 mm/h intensity and 25 m^2 of impervious area connected to 1 m^3 of soakaway. The total simulation time is 10 days (or until the soakaway has emptied). The soakaway is assumed to be empty at the start of the inflow event.

In order to understand the effect of considering reduced infiltration rates (Eq. (3)), the six different base scenarios were also run using constant infiltration rates $f = 1$(Eq. (3)).

2.4.2. Rainfall time series scenarios

The rainfall time series scenarios evaluate the performance of the new model under a continuous simulation. Instead of using a constant inflow hydrograph as it was done in the base scenarios, the inflow to the soakaway was derived from a 1-month rainfall time series (19th of July to 18th of August 2005) collected at Rødovre Waterworks, Copenhagen by the Danish Meteorological Institute (see Jørgensen et al., 1998 for details about measurements). The rainfall time series was used to compute the runoff from the impervious area which is then diverted into the soakaway. A ratio of 1:25 m^3/m^2 of soakaway volume versus impermeable area was chosen. The ratio was selected to demonstrate the performance of the algorithm and so avoids overflow situations for the given scenario, while still resulting in reasonably big water level variations in the soakaway. For engineering purposes other ratios may be needed depending on the soil type and accepted overflow frequency. Initial losses of the impervious area were neglected; this means that all the rainfall falling onto the impervious area is diverted into the soakaway. The time of concentration of the impervious area is expected to be negligible so no time delay between rainfall and runoff was assumed.

2.4.3. Impact of parameter variation

To evaluate the impact of parameter variation, some additional scenarios have been run modifying the initial groundwater level and saturated hydraulic conductivity. Two variables have been selected as representative parameters for these results – the emptying time of soakaway and the maximum groundwater mounding depth 5 m away from the soakaway center within the first 10 days of simulation. The parameters varied are shown in Table 4. The inflow hydrograph was calculated using Eq. (18) for all these scenarios.

2.5. Mass balance

The mass balance of the new model was calculated at each time step by summing the water volume in the soakaway, in the unsaturated zone and in the groundwater mound. Water in the unsaturated zone was estimated by subtracting the groundwater saturated zone recharge volume from the volume infiltrated from the soakaway. The length of the soil domain, L, was extended to 200 m for these calculations, in order to be able to assume no-flow conditions at the boundary $x = L$.

Mass balances were also checked for the two-dimensional model simulations. The water in the soil was calculated by integrating the moisture content over the soil area, the flow into the soil was set to be equal to the outflow from the soakaway, and the flow out of the soil domain was calculated by integrating the flux on the right boundary of the soil.

3. Results and discussion

3.1. Base scenarios

Fig. 4a–c shows the soakaway water depth as calculated by the new model and the two-dimensional model for the sandy clay loam (4a), silt loam (4b) and clay loam (4c).

The graphs show that the new model is able to account for the effects of a shallow water table, and that there is a clear difference between the results of shallow and deep groundwater conditions. However, the water level during the emptying phase is slightly overestimated for most conditions shown above (except for the infiltration rate for sandy soils under the influence of a shallow groundwater table), which means that the infiltration rate is somewhat underestimated for a majority of the scenarios.

Fig. 5a–c shows the groundwater mound profiles at various times after the inflow start for the three soil types. Only results for shallow groundwater conditions are shown here since the mounding is very small or even non-existent for the deep groundwater scenarios. Mounding depths in the two-dimensional model were taken from the elevation of the $h = 0$ line.

The graphs show that the new model underestimates the mounding depth close to the soakaway, and generally overestimates it at distances further away from the soakaway. Carleton (2010) observed similar behavious, reporting an underestimation of 15% of the maximum height of a groundwater mound, however a direct comparison with his results is not possible due to differences in the model domain and his assumption of a constant infiltration rate. One reason for the underestimation is that the mound in the two-dimensional model is connected to the soakaway water table, and thus the mounding depths close to the side of the soakaway will be similar to the elevation of the water level in the soakaway. In the new model, the mound grows upwards from the initial water table level, and is thus not directly connected to the soakaway water level. Moreover, the groundwater flow is assumed to be horizontal only in the new model, but in the two-dimensional model the main flow direction close to the center of the mound is vertical.

Another reason for the difference in mounding depths between the two models is due to the specific yield which is constant in the new model whereas it is a function of elevation in the two-dimensional model. Mass balances of the water in the groundwater mound have been approximated by multiplying the area underneath the groundwater mound with the soil porosity (θ_s). The mass balances show that the groundwater mound is generally underestimated in the initial period after the start of inflow event by the new model, but is later overestimated. The differences between models relate to water in the unsaturated zone. The new model overestimates the initial soil moisture deficit (when the mound is small and close to the initial water table) and underestimates the moisture deficit for larger mounds. Fig. 6 shows the water volume underneath the groundwater mounds at various times for the two models and three soil types.

To illustrate the importance of taking into account the reduction of infiltration rates due to the distance between the bottom of the soakaway and the top of the mounding, Table 5 shows a comparison of results for the base scenario simulations with a constant infiltration rate and with the new approach where infiltration rates are adjusted every time step using Eq. (3). There are significant differences in the model outputs for shallow groundwater table, whereas almost no differences are observed for the deep groundwater table. The small difference for deeper mounds is because the mounding depth for the deep groundwater case did not increase enough to affect infiltration rates. Results show that

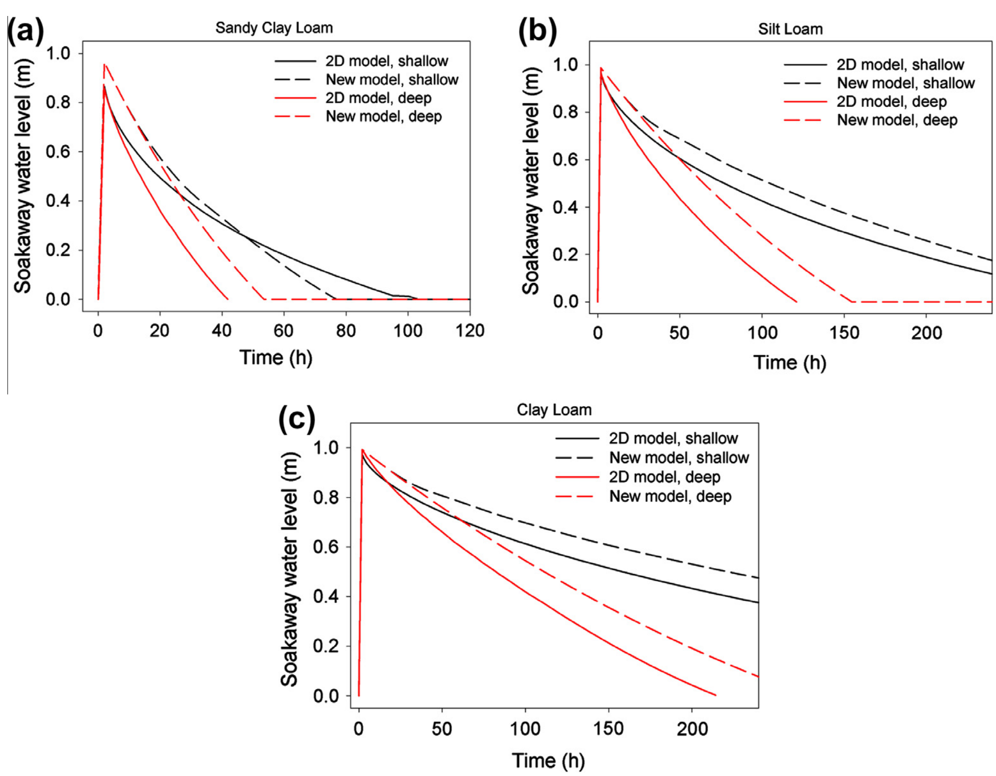

Fig. 4. (a–c) Soakaway water depth as a function of time for (a) sandy clay loam, (b) silt loam and (c) clay loam under shallow and deep groundwater conditions. The solid line shows the results from the two-dimensional model (representing "real" conditions), and the dashed line results from the new model presented in this paper. The black color denotes shallow groundwater conditions and the red color deep groundwater conditions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

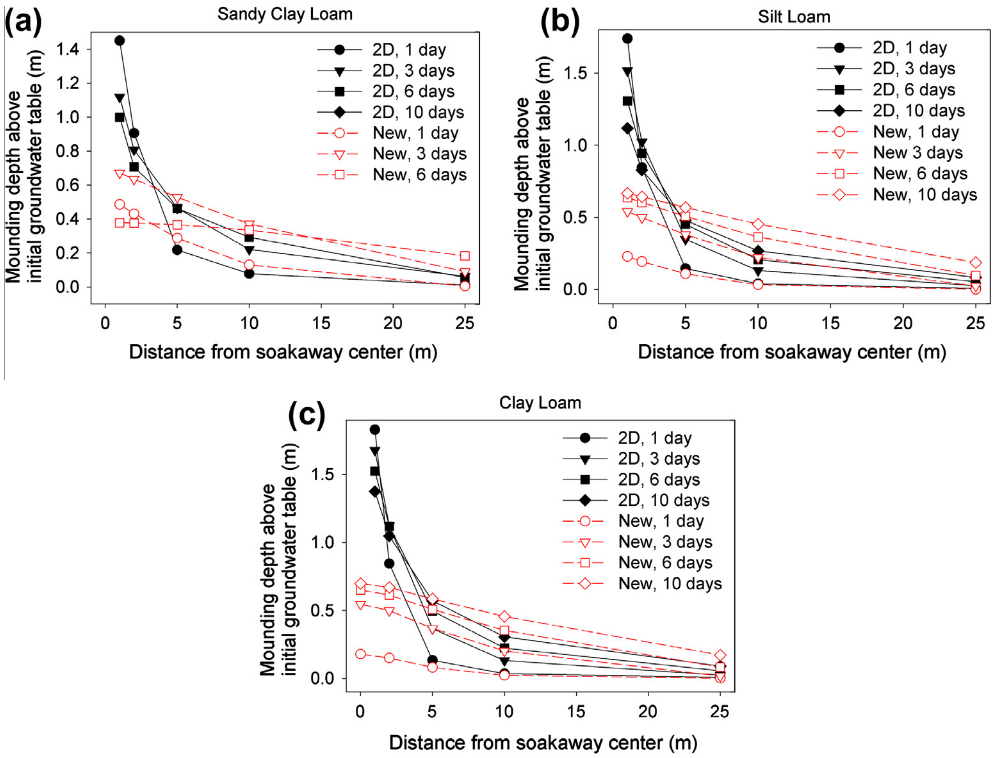

Fig. 5. (a–c) Groundwater mounding profiles at 1 day, 3 days, 6 days and 10 days after the start of the inflow event for sandy clay loam (a), silt loam (b) and clay loam (c) with shallow groundwater conditions.

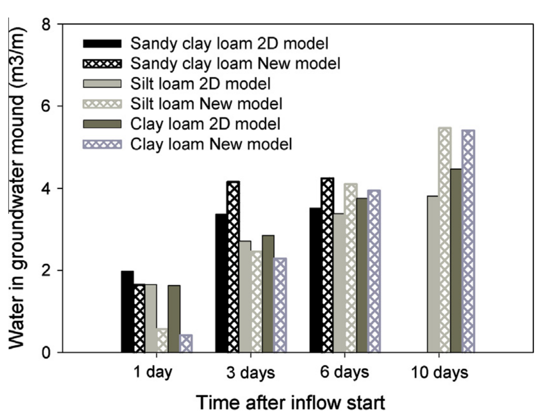

Fig. 6. Total volume of water in the groundwater mound at various times after the start of the inflow event for the two models and three soil types.

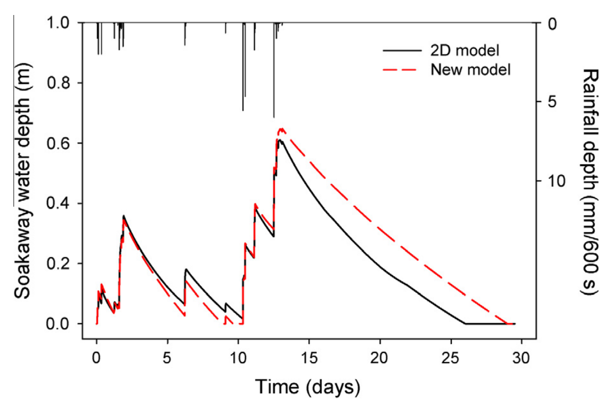

Fig. 7. Soakaway water level calculated by the two-dimensional model (solid black line) and the new model (dashed red line) for the silt loam shallow groundwater scenario, using an inflow hydrograph based on a rainfall time series from July to August, 2005. The rainfall hyetograph is shown at the top of the figure, right y-axis. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Results obtained for the Base scenarios by considering constant infiltration rates $f = 1$. The results are shown as a comparison with the Base scenarios where infiltration rates were updated every time step according to Eq. (3).

Scenario	Sandy clay loam	Silt loam	Clay loam	
Shallow	Reduction in the emptying time [%]	31	54	64
	Increase in the maximum mounding height [%]	24	60	78
Deep	Reduction in the emptying time [%]	0.0	0.0	1.0
	Increase in the maximum mounding height [%]	0.0	0.0	1.1

if the reduction in infiltration rate due to mounding is not considered, then the emptying time of the soakaway would be underestimated and the mounding overestimated. These effects are most pronounced for clay loam and less for sandy clay loam.

3.2. Rainfall time series results

Fig. 7 shows the soakaway water level for the rainfall time series scenario, using the silt loam soil with shallow groundwater conditions and Fig. 8 shows the groundwater mounding time series for some selected distances away from the soakaway center.

The results are similar to the base scenarios in that the soakaway water level is generally well represented by the new model, whereas groundwater mounding depths show a larger deviation. Fig. 8 shows, just like the groundwater results from the base

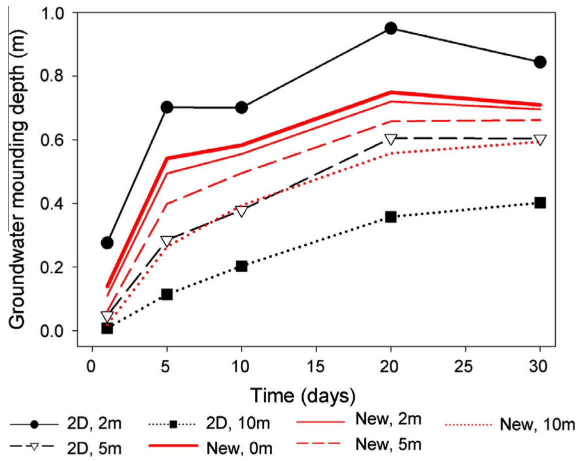

Fig. 8. Groundwater mounding depth at selected times, 0 m, 5 m and 10 m from the soakaway center, calculated by the two-dimensional model (black line) and the new model (red line) for the silt loam shallow groundwater scenario, using an inflow hydrograph based on a rainfall time series from July to August, 2005. Mounding depths for the two-dimensional model below the soakaway center are omitted since they equal the water level in the soakaway. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

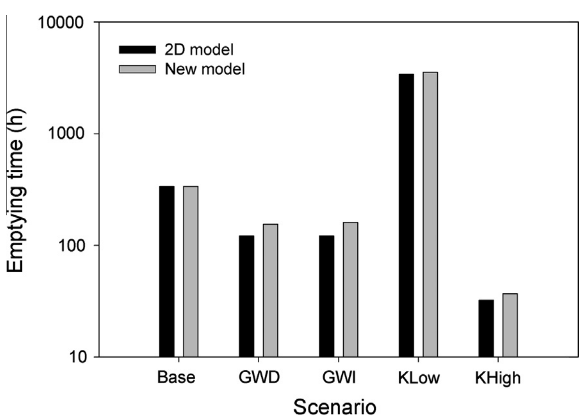

Fig. 9. Emptying times for the different scenarios. For details on each scenario, refer to Table 4. Note that the scale of the *y*-axis is logarithmic due to the large variations in emptying times.

scenario, that the new model underestimates the mounding depth close to the soakaway, and overestimates it further away. We expect these trends to be general for all cases.

3.3. Impact of parameter variation

The results of the alternative scenarios and the base scenario are summarized in Fig. 9 (emptying time) and Fig. 10 (maximum groundwater mound).

The general pattern seen in Fig. 9 is that the new model tends to overestimate the emptying time slightly – on average the error is 13% overestimation compared to the emptying time for the two-dimensional model. The deviation is particularly pronounced for the deep and intermediate groundwater scenarios, which is similar to the results shown in Fig. 4a–c. If only the four shallow groundwater scenarios (i.e. all but the deep and intermediate groundwater scenarios) are considered, the average deviation is only 5%. These differences should be viewed in the light of other uncertainties, such as the uncertainty associated with the soil moisture retention parameters and uncertainties in rainfall measurements and runoff calculations of the soakaway catchment. The largest error is seen for the deep groundwater scenario, where the

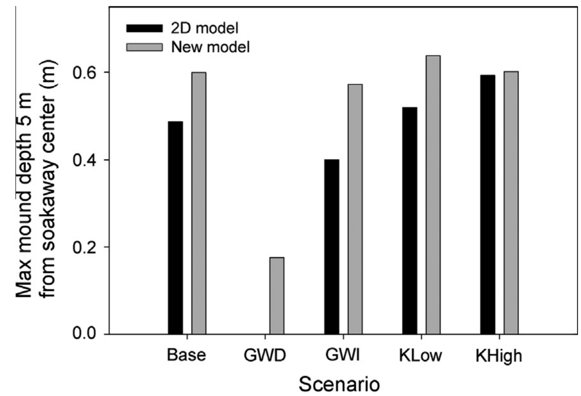

Fig. 10. Maximum groundwater mound at 5 m from soakaway center (within the first 10 days after the inflow start) for the different scenarios. For details on each scenario, refer to Table 4.

one-dimensional model overestimates the emptying time by 32%. The same increase in emptying time would be obtained if the two-dimensional model was run with a K_s of 1.67×10^{-6} m/s instead of 1.25×10^{-6} m/s, i.e. if the estimate of K_s was 25% lower than the actual K_s. The error that the simplified model introduces is therefore likely to be much smaller than errors resulting from uncertainty of input parameters.

Large differences in maximum mounding depth are seen for the deep and intermediate groundwater scenarios (Fig. 10). This is likely to be a result of the new model underestimating the traveling time of the wetting front. For the deep groundwater scenario, the wetting front in the two-dimensional model does not reach the groundwater table before the simulation ends. For the intermediate groundwater scenario, the wetting front reaches the groundwater table in the two-dimensional model but at a later stage compared to the one-dimensional model which leads to the groundwater mounding depth being overestimated by the new model.

For the scenarios with a shallow water table the new model tends to overestimate the mounding depth. The average deviation of the new model results relative to the two-dimensional model results for the four shallow groundwater scenarios in Fig. 10 is 18%.

3.4. Mass balance

The maximum mass balance error for the new model was found to be less than 1% for all base scenarios and the rainfall time series scenario with silt loam soil and shallow groundwater. The average error over the entire simulation period was around 0.2%. The highest relative errors were observed after the start of groundwater recharge, i.e. when the total inflow volumes were relatively small. The error decreases by reducing the time step, which is expected because of the sequential approach used to solve the infiltration equation (Eq. (4)) and the groundwater mounding equation (Eq. (8)) in each time step. An iterative solving method would reduce the mass balance error, but would also increase the computational cost. Given that the current errors are small, the sequential method employed here is considered to be satisfactory.

The mass balance error of the two-dimensional model was also found to be small (around 1%). The error was primarily related to the spatial discretization, and test runs with a finer mesh resulted in reduced mass balance errors.

3.5. Suitability of models in an integrated environment

3.5.1. Computational time

The new model simulates a 10-day period in approximately 3 s, compared to an average of 30 min runtime for the two-dimensional model on the same computer, i.e. the new model is around 600 times faster. The computational time of the new model is relatively independent of soil type, initial conditions and inflow pattern, the runtimes for the different scenarios above varied between 3.3 and 3.6 s. The two-dimensional model runtime depends a lot on the initial conditions and the soil type and whether the soakaway empties within the simulated period or not. The fastest simulation (clay loam, shallow groundwater conditions) took a couple of minutes to run, and the slowest (sandy clay loam, deep groundwater conditions) several hours. The very long simulation times were mainly observed in situations where the soakaway emptied before the end of the 10-day period. Thus, this large difference in computational time may be due to a substantial reduction in time step length when the soil is drying and the hydraulic conductivity in the unsaturated soil becomes very small.

Reducing the time step from 600 s to 60 s in the new model resulted in approximately 10 times longer runtimes, and no significant difference in results and mass balance error. The maximum mass balance error was found to be less than 1% for all base scenarios with a 600 s time step length. Increasing it to 3600 s reduced the runtime by a factor 6, and some marginal differences were seen in the results, with maximum mass balance errors up to 2–3%. Thus, the magnitude of the current time step length seems to be appropriate.

3.5.2. Data input

Both models require essentially the same input data: soakaway geometry, depth to groundwater and total aquifer depth, and parameters of the soil moisture retention curve. Thus, if the new model is used in combination with an existing distributed groundwater model that contains these parameters, it requires no additional calibration.

3.5.3. General suitability

While the two-dimensional model produces more accurate results, it is time demanding to set up and requires significant computational effort. This makes it unsuitable to use in an integrated modeling environment if the effect of soakaways are to be assessed on a larger scale than just one or a few individual allotments. The new model is several orders of magnitude faster and the results for the soakaway water level are reasonably accurate (5–10% error relative to the two-dimensional model). It can also provide estimates of local groundwater mounding effects within a 10–20% error margin which is useful if it is coupled to a groundwater model with a coarse spatial resolution. Thus we believe that it can be a valuable tool to use in integrated modeling studies involving both urban drainage and groundwater on larger scales.

4. Conclusions

We have developed a new modeling approach for soakaway infiltration in the presence of a shallow groundwater table that is able to account for the full coupling of the groundwater level and infiltration rate, as well as estimating the local groundwater mounding. The variation of infiltration rates due to the distance between the top of the mounding and the bottom of the soakaway has been shown to have a significant effect on the output results. The model is intended to improve integrated modeling of urban drainage, soakaways and groundwater since it can account for local groundwater mounding effects even where the spatial resolution of the groundwater model is too coarse for this.

The errors introduced by the simplifications in the new model, are generally within 10–20% of the results produced by a two-dimensional unsaturated/saturated flow model based on Richards' equation. The errors are smaller when the groundwater table is shallow, and thus the new model is most suited for use when the infiltration rate is influenced by interaction with the groundwater. Mass balance errors of the new model are less than 1% for all tested scenarios. The new model is based on the same input parameters as the two-dimensional model and has runtimes several orders of magnitude faster than the two-dimensional model (on average 600 times faster). This makes coupled groundwater-urban water modeling feasible. Computational times for a full description of urban water flows, with a saturated zone (groundwater) and unsaturated zone are otherwise too high to make coupled groundwater-urban water modeling possible.

An obvious limitation with the current model setup is that it only considers one individual soakaway. To be useful in an applied context, it must be further developed to account for multiple soakaways as well as the interaction between neighbouring groundwater mounds. A solution for the Hantush model that is similar to the one used in this paper, but can be used for any number of recharge basins, is proposed by Manglik et al. (2004). It would also be of interest to investigate possible upscaling or aggregation methods for soakaways connected to a groundwater model (see e.g. Roldin et al., 2012a). Another topic for future research is to apply the new model to a case study, in order to test it in a more applied context.

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